Aravindraj M

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CAREER OBJECTIVE

B.Tech student in Artificial Intelligence and Data Science with hands-on experience in software development, AI agents, REST APIs, and data-driven applications. Strong foundation in Python, C++, DSA, OOP, and SQL.

TECHNICAL SKILLS

Languages: Python, C++, C, SQL

Core Concepts: DSA (150+ LeetCode Problems), OOP, DBMS

Software Development: REST APIs, FastAPI, Git, GitHub, Docker

Web: HTML, CSS, JavaScript, React

Databases: MySQL, SQLite, Supabase,MongoDB

AI/ML: Machine Learning, Deep Learning,Gen AI,Retrieval-Augmented Generation (RAG),LangChain,LangGraph

Computer Vision: OpenCV, MediaPipe, YOLO,CNN

Experience

AI/ML Member — iQube Innovation Center, KCT | 2025 – Present
- Develop AI-powered applications using AI Agents, RAG, LLMs, ML, DL, and Computer Vision.
- Build AI pipelines and backend/API integrations while collaborating on research, prototyping, and end-to-end project development.

PROJECTS

Enterprise AI Agent — RAG-Powered Knowledge Assistant, *2026*
- Built a LangGraph-based AI agent with tool calling to route user queries across retrieval and application-specific tools.
- Implemented hybrid search combining BM25 and vector search for context retrieval before LLM response generation.
- Integrated the agent with FastAPI, MongoDB Atlas, and React, developing the supporting ingestion, embedding, and REST API pipelines.

Tech Stack: *Python, LangGraph, FastAPI, React, MongoDB Atlas, BM25, Sentence Transformers, Groq, REST APIs*

PatientPath AI – Hospital Flow Optimization , *2026*
- Built an AI-driven workflow platform using **YOLO and OpenCV** for real-time occupancy monitoring.
- Implemented ML models for **wait-time prediction, bottleneck detection, and staff allocation**.
- Developed a **FastAPI backend** with real-time dashboards, risk scoring, and automated alerts.

Tech Stack: *Python, FastAPI, YOLO, XGBoost, OpenCV, MySQL, n8n*

Bharatanatyam Mudra Detection , *2025*
- Developed an AI-powered Bharatanatyam Mudra Recognition System using MediaPipe and YOLO for real-time gesture recognition.
- Trained deep learning models on a custom dataset for single- and double-hand mudra classification.
- Designed interactive learning modules including quizzes and flashcards for real-time user feedback.

Tech Stack*:* *Python, OpenCV, MediaPipe, TensorFlow, YOLO*

EDUCATION

B.Tech – Artificial Intelligence and Data Science, *Kumaraguru College of Technology, Coimbatore* 2024 – Present

ACHIEVEMENTS

Runner-up – Maatram Hackforge 2.0 Hackathon

Finalist – Sankara Innovation Challenge 2026